EXP 9 ADC

```
clc;
clear;
close all;

% 1. PARAMETERS
fs = 100000;                    % Sampling frequency
t = 0:1/fs:0.02;                % Time vector
fm = 500;                       % Message frequency (Hz)
fc = 5000;                      % Carrier frequency (Hz)
SNR = 10;                       % Signal-to-noise ratio (dB)
mu = 0.8;                       % Modulation index

% 2. MESSAGE SIGNAL
m = cos(2*pi*fm*t);

% 3. CARRIER SIGNAL
c = cos(2*pi*fc*t);

% 4. AM MODULATION
am = (1 + mu*m).*c;

% 5. ADD AWGN NOISE
signalPower = mean(am.^2);
noisePower = signalPower / (10^(SNR/10));
noise = sqrt(noisePower)*randn(size(am));
noisy = am + noise;

% 6. ENVELOPE DETECTION
rectified = abs(noisy);

% Low-pass filter using moving average
windowSize = round(fs/fc);
filterCoeff = ones(1,windowSize)/windowSize;
env = conv(rectified, filterCoeff, 'same');

% 7. RECOVER MESSAGE SIGNAL
recovered = (env - mean(env))/mu;

% 8. PLOT ALL SIGNALS
figure('Name','AM Modulation and Demodulation');

```

```
subplot(5,1,1);
plot(t,m,'b');
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

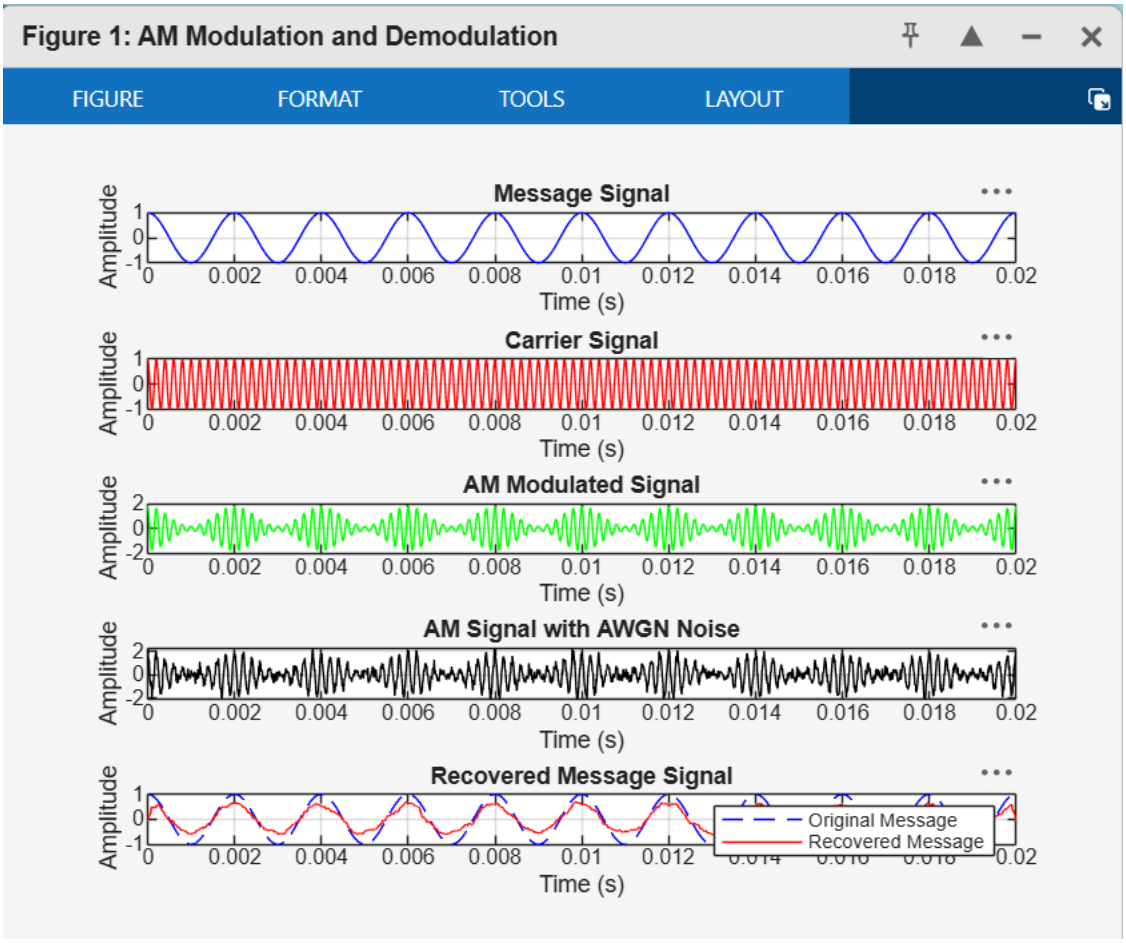
subplot(5,1,2);
plot(t,c,'r');
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,3);
plot(t,am,'g');
title('AM Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;


subplot(5,1,4);
plot(t,noisy,'k');
title('AM Signal with AWGN Noise');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,5);
plot(t,m,'b--');
hold on;
plot(t,recovered,'r');
title('Recovered Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
legend('Original Message','Recovered Message');
grid on;

% 9. DISPLAY COMPLETION
disp('AM modulation and demodulation completed successfully.');
```


Figure 1: AM Modulation and Demodulation
FIGURE
FORMAT
TOOLS
LAYOUT
Message Signal
Carrier Signal
AM Modulated Signal
AM Signal with AWGN Noise
Recovered Message Signal
Amplitude
Time (s)
0
0.002
0.004
0.006
0.008
0.01
0.012
0.014
0.016
0.018
0.02
Original Message
Recovered Message